A Decay-Safe Gauged $U(1)$ Origin of the $\mathbb{Z}_{17}$ Spin(10) Axion Model

Minimal Complete-Pair Anomalons and Mass-Dependent Two-Loop Audit

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2 August 2026

Abstract

In the SO(10)×U(1)$_{\rm PQ}$ axion model of Ernst, Ringwald and Tamarit, gauging $\mathbb{Z}_{17} = U(1)_{\rm PQ}$ mod 17 requires anomaly-cancelling matter. For arbitrary vector-like $16 \oplus \overline{16}$ spectator pairs massed by the PQ singlet S, the mixed anomaly alone forces $k \equiv 5 \pmod{17}$, independent of their individual charges. Within the identical-pair ansatz all discrete anomaly classes then select the unique charge residues $\{2, 11\} \pmod{17}$. The minimal completion has five pairs, shifts the gluonic coefficient to $2N = -68$, and gives $N_{\rm DW}^{\rm cover} = 17$, $f_a = v_S/17$, unit physical domain-wall number and $E/N = 8/3$. The benchmark $v_S = M_I = 6.31 \times 10^{11}\,\text{GeV}$, $y_s = \sqrt{2}$ predicts $m_a = 153.5 \pm 2.0\,\mu\text{eV}$, $\nu = 37.11 \pm 0.49$ GHz and $g_{a\gamma\gamma} = (2.335 \pm 0.125) \times 10^{-14}\,\text{GeV}^{-1}$.

At the discrete-EFT level we distinguish a local PQ-breaking operator from a spurion set that can close into a vacuum amplitude. Although $\mathbb{Z}_{17}$ permits fermionic operators of dimensions 6, 9, 10 and 12, the renormalizable theory preserves an exact spectator vector number V. An intentionally over-complete enumeration of all monomials satisfying Lorentz parity, the Spin(10) centre and $\mathbb{Z}_{17}$ excludes a PQ-sensitive, $V = 0$ closure through $P = \sum_i (d_i - 4) = 11$. The bound is saturated at $P = 12$ by four dimension-6 portals plus one PQ-conserving dimension-8 vector-breaking spurion. An explicit 32-dimensional Clifford-algebra and Grassmann audit finds nonzero Spin(10) and Lorentz contractions, verifies Fermi statistics, and identifies a connected two-loop graph. Wilsonian NDA gives $|\Delta\bar{\theta}| \lesssim 9.04 \times 10^{-28} |C_{\rm eff}|$. The extra $\mathbb{Z}_3$ proposed in v15 is unnecessary and is removed.

We then construct a decay-safe anomaly-free origin of the discrete symmetry. Gauge $U(1)_X$, break it with a PQ-neutral scalar Φ_{17}, and add three complete pairs with charges $(1, 16)$, $(14, 3)$ and $(1, -18)$. They contribute $(+34, +272, +16592)$ to the mixed Spin(10), gravitational and cubic anomalies, exactly cancelling the light sector, while their accidental-PQ anomaly vanishes. Each pair has a renormalizable portal to ordinary 16_F matter through an already-present 10_H or S. Mixed-anomaly cancellation forces the number of single-Φ-massed pairs to be odd; one pair would require roots of $t^2 - 17t + 76$, whose discriminant is -15. Thus three is minimal in this complete-pair-only ansatz (with no Spin(10)-singlet anomalons), and exact enumeration finds the displayed triple uniquely in the stated existing-field portal basis.

The decay portals change, rather than invalidate, the quality audit. A heavy-inclusive necessary-condition enumeration excludes a vacuum closure through $P = 7$ and is saturated at $P = 8$ by four dimension-five portals plus one dimension-eight spectator spurion. An explicit nonzero connected two-loop graph produces $\Phi^4 (S^\dagger)^{18} (10_H^\dagger)^2$. Its repeated-pole integral retains all three masses and gives $2|A_8|/\chi = 6.04 \times 10^{-47} |C_8|$ at $v_\Phi = 10^{17}\,\text{GeV}$. The direct dimension-21 scalar operator remains the largest explicitly computed unit-coefficient term, $2|A|/\chi = 4.60 \times 10^{-43}$. A normalized decay portal as small as 3.1×10^{-20} gives a lifetime below one second; every 16 component has a nonzero 10_H channel in the explicit Clifford representation. The construction is anomaly-free as an effective field theory, not a string completion. Perturbativity to the reduced Planck scale is not established: the present one-loop Spin(10) envelope reaches a Landau pole below that scale. Flavour/Clebsch fits, hierarchy stabilization, the full scalar vacuum and the string network remain open. Three

fail-closed engines perform 65, 59 and 42 checks; the release adds independent charge, decay and loop-quadrature tests.

1 Model and conventions

Fields and PQ charges: 16_F (+1, three generations), 10_H (−2), $\overline{126}_H$ (−2), 210_H (0), S (+4) [1]; the axion is intended to be dominantly $\arg S$. The previously proposed 54-channel $\overline{126}_H^2 10_H^2 S^2$ locking term vanishes on the selected $\overline{126}_H$ direction, so the complete phase Hessian remains open. At low energy we impose $\mathbb{Z}_{17} := \mathrm{U}(1)_{\mathrm{PQ}} \bmod 17$; Section 5 realizes it as the remnant of a primitive gauged $U(1)_X$. Spectators: k identical $16_s \oplus \overline{16}_s$ pairs with $\mathbb{Z}_{17}$ charges (a, b), masses from $S\,16_s\overline{16}_s$, i.e. thresholds $M_s = y_s v_S/\sqrt{2}$. Discrete-gauge protection of axions has a long history [6, 7, 8, 9]; the specific result below concerns this particular $\mathbb{Z}_{17}$ extension of the ERT model.

2 The theorem

Theorem 1 (minimality and residue uniqueness, identical-pair ansatz)**.** *The $\mathbb{Z}_{17}$ anomaly classes (mixed $SO(10)^2$, gravitational, cubic [24, 2]) all vanish modulo 17 if and only if $k \equiv 5 \pmod{17}$ and $\{a, b\} \equiv \{2, 11\} \pmod{17}$. The minimal solution therefore contains five pairs, and the charge* residues *are unique up to interchange; $(2, -6)$ is one integer lift.*

Proof. The mass operator forces $a + b \equiv -4 \equiv 13$. The mixed class is $6 + 2k(a{+}b) \equiv 6 - 8k$; with $8^{-1} \equiv 15$, vanishing requires $k \equiv 90 \equiv 5$, i.e. $k = 5 + 17m$. The gravitational class $48 + 16k(a{+}b) \equiv 14 + 4k$ vanishes automatically on the same set. The cubic class uses $a^3 + b^3 = (a{+}b)^3 - 3ab(a{+}b) \equiv 4 - 5ab$, giving $48 + 80(a^3{+}b^3) \equiv 11 - 9ab \equiv 0$, i.e. $ab \equiv 5$. The residue pair is then the root set of $x^2 - 13x + 5 \equiv 0 \pmod{17}$, whose only roots are 2 and 11; an exhaustive residue scan confirms uniqueness. □

The five-pair lower bound is stronger than the theorem's identical-pair wording. For arbitrary residues $(a_i, 13 - a_i)$, the mixed class is

$$6 + 2\sum_{i=1}^{k}\big[a_i + (13 - a_i)\big] \equiv 6 + 9k \pmod{17}, \tag{1}$$

so every charge assignment requires $k \equiv 5 \pmod{17}$. Direct enumeration gives no solution for $k \leq 4$ and 83 232 ordered cubic solutions among the 17^5 tuples at $k = 5$. Thus pair-count minimality is charge-independent, whereas the residue uniqueness $\{2, 11\}$ is specifically an identical-pair result. The general cubic class used here is $48 + 16\sum_i[a_i^3 + (13 - a_i)^3]$; the alternate engine incorrectly used 80 per pair (valid only after five identical pairs), although its solution count happened to remain unchanged.

The five pairs shift the gluonic coefficient to $2N = 2[6 + 10(a{+}b)] = -68$, so

$$N_{\mathrm{DW}}^{\mathrm{cover}} = \frac{|2N|}{X_S} = \frac{68}{4} = 17, \qquad f_a \longrightarrow \frac{v_S}{17}, \tag{2}$$

in the discrete EFT, or equivalently the $v_\Phi \gg v_S$ limit of Section 5; the exact continuous-theory projection is given in Eq. (19). The ratio $E/N = 8/3$ is preserved (complete SO(10) multiplets). All seventeen covering-space minima form one gauge orbit ($4 \times 13 \equiv 1 \bmod 17$): the physical domain-wall number is one, and the one-wall object is the string sector $(\ell, n) = (13, -3)$ with S-winding $2\pi/17$ (we reserve k for the pair multiplicity).

3 Benchmark and inputs

One-loop running of $\alpha_1^{-1}(M_Z) = 59.02$ (GUT-normalised), $\alpha_2^{-1}(M_Z) = 29.57$, $\alpha_s(M_Z) = 0.1179$ with 2HDM coefficients $(21/5, -3, -7)$ below M_I, Pati–Salam coefficients $(-7/3, 2, 26/3)$ above it, and matching $\alpha_{2R}^{-1} = \frac{5}{3}\alpha_1^{-1} - \frac{2}{3}\alpha_4^{-1}$ gives $M_I = 6.31 \times 10^{11}\,\mathrm{GeV}$, $M_{\mathrm{GUT}} = 9.92 \times 10^{15}\,\mathrm{GeV}$, $\alpha_{\mathrm{GUT}}^{-1} = 37.31$. *Gauge unification does not fix the singlet scale [1]; we separately impose the benchmark identifications $v_S = M_I$ and $y_s = \sqrt{2}$* (so $M_s = M_I$). Then, with $\chi^{1/4} = 75.5(5)$ MeV and the QCD photon coefficient 1.92(4) from Grilli di Cortona et al. [20],

$$m_a = \frac{\chi^{1/2}}{f_a} = 153.5 \pm 2.0\,\mu\mathrm{eV}, \qquad \nu = \frac{m_a}{h} = (37.11 \pm 0.49)\ \mathrm{GHz}, \tag{3}$$

$$g_{a\gamma\gamma} = \frac{\alpha}{2\pi f_a}\left|\frac{E}{N} - 1.92\right| = (2.335 \pm 0.125) \times 10^{-14}\,\mathrm{GeV}^{-1}, \tag{4}$$

where the mass and frequency uncertainties propagate the susceptibility error and the coupling uncertainty propagates the 1.92(4) coefficient. The compatible determination $\chi^{1/4} = 75.6(1.8)(0.9)$ MeV [19] would widen the mass uncertainty to about $8\,\mu$eV. The spectator shift is $\Delta\alpha_{\mathrm{GUT}}^{-1} = -\frac{40/3}{2\pi}\ln(M_{\mathrm{GUT}}/M_s) = -20.5$ at the benchmark y_s; other y_s values shift it (e.g. -26.1 at $y_s = 0.1$), so y_s is an explicit input of the proton estimate.

4 Axion quality: local operators and vacuum closure

At the reduced Planck scale we assume every local operator allowed by the gauge group is present, with a dimensionless Wilson coefficient of order unity, and that no intermediate threshold carries additional explicit PQ breaking. This assumption is essential: heavy fields can modify naive Planck power counting [11].

Scalar sector and the local-operator trap

The 10_H and $\overline{126}_H$ tensors have an odd number of SO(10) vector indices, whereas 210_H has an even number. Deltas and the rank-ten epsilon tensor contract an even total number of vector indices, so an invariant contains an even number of odd-index tensors. Including S (charge $+4$), every scalar invariant therefore has $Q_{\mathrm{PQ}} = 0 \pmod 4$. Combining this with $\mathbb{Z}_{17}$ gives $|Q_{\mathrm{PQ}}| \geq \mathrm{lcm}(4, 17) = 68$, saturated by $S^{17}/M_{\mathrm{Pl}}^{13}$. At the benchmark, the one-sided coefficient is $(v_S/\sqrt{2})^{17}/(M_{\mathrm{Pl}}^{13}\chi) = 3.24 \times 10^{-37}$. With $V = Ae^{ia/f_a} + \mathrm{h.c.}$ the worst-phase shift is

$$\frac{|\Delta\bar{\theta}|_{S^{17}}}{|c_{17}|} = \frac{2(v_S/\sqrt{2})^{17}}{M_{\mathrm{Pl}}^{13}\chi} = 6.47 \times 10^{-37}, \qquad M_{\mathrm{Pl}} = 2.435 \times 10^{18}\,\mathrm{GeV}. \tag{5}$$

Fermions invalidate any proof that stops there; multi-fermion operators are known to matter in axion-quality analyses [10]. Write $F = 16_F$, $s = 16_s$ and $b = \overline{16}_s$, with $Q_{\mathrm{PQ}} = (1, 2, -6)$. SO(10)$\times\mathbb{Z}_{17}$ permits, among others,

$$\begin{aligned}
\mathcal{O}_6^- &= \frac{(S^\dagger)^3(Fb)_1}{M_{\mathrm{Pl}}^2}, && (d, Q, V) = (6, -17, -1),\\
\mathcal{O}_8^0 &= \frac{(S^\dagger)^2[(ss)_{10}]^2}{M_{\mathrm{Pl}}^4}, && (d, Q, V) = (8, 0, +4),\\
\mathcal{O}_9 &= \frac{(S^\dagger)^5(bb)_{10}10_H}{M_{\mathrm{Pl}}^5}, && (d, Q, V) = (9, -34, -2),\\
\mathcal{O}_{10} &= \frac{10_H^\dagger[(bb)_{10}]^3}{M_{\mathrm{Pl}}^6}, && (d, Q, V) = (10, -34, -6),\\
\mathcal{O}_{12} &= \frac{(S^\dagger)^6(Fs)_{10}(ss)_{10}}{M_{\mathrm{Pl}}^8}, && (d, Q, V) = (12, -17, +3),
\end{aligned} \tag{6}$$

where $V(s) = +1$, $V(b) = -1$ and all ERT fields have $V = 0$. The representation products used in these explicit contractions are standard, in particular $16 \otimes 16 = 10_s \oplus 120_a \oplus 126_s$ [12]. The dimension-10 and -12 terms are regression cases missed by a bilinear-only enumeration.

Nor does $\mathcal{O}_9$ by itself generate the previously claimed term linear in its Majorana insertion. For one Dirac pair,

$$\mathcal{M} = \begin{pmatrix} 0 & M_s \\ M_s & \delta e^{i\phi} \end{pmatrix}, \quad \mathrm{tr}(\mathcal{M}^\dagger \mathcal{M}) = 2M_s^2 + |\delta|^2, \quad \det(\mathcal{M}^\dagger \mathcal{M}) = M_s^4. \tag{7}$$

Both singular values, hence the one-loop Coleman–Weinberg potential, are independent of ϕ. A local PQ-breaking operator is therefore not automatically a one-insertion axion potential.

Finite multi-spurion proof

The renormalizable Lagrangian preserves the spectator vector number $U(1)_V$: the mass Ssb is neutral and there is no dimension-four or lower $Q_{\mathrm{PQ}} = 0$, $V \neq 0$ interaction. Consequently a vacuum graph must have total $V = 0$. Each Planck insertion has $Q_{\mathrm{PQ}} = 17m$; after its fermions are closed with PQ-conserving vertices, scalar index parity also requires total $Q_{\mathrm{PQ}} = 0 \pmod 4$. A PQ-sensitive target thus has

$$\sum_i V_i = 0, \qquad \sum_i m_i \neq 0, \qquad \sum_i m_i = 0 \pmod 4. \tag{8}$$

The supplied engine enumerates a deliberate *superset* of local monomials: F, s, b and all conjugates; $S, S^\dagger$; charge-∓ 2 tensor scalars; even Weyl count; trivial Spin(10)-centre charge; and $Q_{\mathrm{PQ}} = 0 \pmod{17}$. It allows mixed-chirality contractions that may not exist, so failure to find a closure is a conservative exclusion. Neutral 210_H, derivatives and gauge strengths can be omitted because they only raise dimension. For

$$P := \sum_i (d_i - 4) \leq 11, \tag{9}$$

every insertion has $d_i \leq 15$; enumeration through dimension 15 is therefore finite and complete for this exclusion. No state satisfying Eq. (8) occurs through $P = 11$. At $P = 12$ the bound is saturated by the actual Spin(10) set

$$4\mathcal{O}_6^- + \mathcal{O}_8^0 : \qquad (Q_{\mathrm{PQ}}, V, P) = (-68, 0, 12). \tag{10}$$

The necessary-condition search was independently reimplemented directly from weak compositions of field multiplicities; it reproduces all twenty low-cost $(P, Q_{\mathrm{PQ}}/17, V)$ frontier classes and the same absence result through $P = 11$ without importing the main enumeration module.

Explicit Spin(10), Lorentz and graph audit

Because spinors are representations of the double cover, this audit uses Spin(10) explicitly ("SO(10)" elsewhere is the standard shorthand). Let Γ^a be a 32×32 Euclidean Clifford representation and $C = \Gamma^2\Gamma^4\Gamma^6\Gamma^8\Gamma^{10}$. On either chiral 16-dimensional subspace, the ten matrices

$$T_\pm^a = (C\Gamma^a)|_{16_\pm}, \qquad (T_\pm^a)^T = T_\pm^a, \qquad \sum_{ij} (T_+^a)_{ij}^* (T_+^b)_{ij} = 16\delta^{ab} \tag{11}$$

are symmetric, so the same-chirality Weyl bilinear $\epsilon_{\alpha\beta} X_i^\alpha (T_\pm^a)_{ij} Y_j^\beta$ has the required Fermi statistics for the 10_s channel. Sparse Grassmann evaluation gives nonzero polynomials for every operator in Eq. (6); in particular $\mathcal{O}_8^0$ has 240 nonzero canonical monomials even when its four s fields share a species.

For the central four-spinor tensor $K_{ijkl} = \sum_a (T^a_+)_{ij}(T^a_+)_{kl}$, the explicit representation gives 640 nonzero entries and

$$\sum_{ijkl} |K_{ijkl}|^2 = 2560, \qquad K^*_{ijkl}(T^1_+)_{ij}(T^1_+)_{kl} = 256, \qquad L_{\text{Lorentz}} = 4. \tag{12}$$

Thus the certificate is not merely a centre-charge candidate: its actual group and Lorentz contractions are nonzero. Four s–b same-chirality propagators (the four s legs may be assigned four of the five spectator copies) and two ordinary-family 10_H-channel chirality flips close the twelve fermions. With five compact vertices and six internal fermion lines, $L = 6 - 5 + 1 = 2$. The resulting scalar operator is proportional to

$$(S^\dagger)^{18}(10^\dagger_H \cdot 10^\dagger_H), \qquad Q_{\text{PQ}} = -68. \tag{13}$$

A special Higgs alignment could make the dot product vanish, which would only strengthen the quality bound by moving the first nonzero vacuum term to higher order.

Treating all masses and scalar insertions as no larger than the matching scale M_s and dropping loop suppressions gives the Wilsonian-NDA upper bound

$$\frac{|\Delta\bar\theta|}{|C_{\text{eff}}|} \lesssim \frac{2M_s^4}{\chi}\left(\frac{M_s}{M_{\text{Pl}}}\right)^{12} = 9.04 \times 10^{-28}. \tag{14}$$

Here C_{eff} absorbs convention-dependent tensor normalisations, Yukawas and flavour factors. For the explicit graph, retaining the two loop factors and using $|v_{10}| \le 246\,\text{GeV}$ yields the diagnostic

$$\frac{|\Delta\bar\theta|_{P=12}}{|C_{\text{eff}}|} \lesssim 9.04 \times 10^{-28}(16\pi^2)^{-2}\left(\frac{246\,\text{GeV}}{M_s}\right)^2 = 5.50 \times 10^{-51}, \tag{15}$$

before any additional flavour suppression. Equation (14), which deliberately omits these small factors, remains the advertised conservative result.

Proposition 1 (quality under the stated EFT assumptions)**.** *At the benchmark $M_s = v_S = 6.31 \times 10^{11}\,\text{GeV}$, the minimal SO(10)×$\mathbb{Z}_{17}$ completion satisfies the neutron-EDM requirement $|\Delta\bar\theta| < 10^{-10}$ provided $|C_{\text{eff}}| < 1.1 \times 10^{17}$. In particular, order-one Wilson coefficients are safe; the leading conservative bound is Eq. (14), not the scalar-only value in Eq. (5).*

5 An anomaly-free continuous origin of $\mathbb{Z}_{17}$

A discrete gauge symmetry can arise as the remnant of a broken continuous gauge group [3, 4, 5]. Introduce a complex Spin(10) singlet Φ with $(X, \text{PQ}) = (17, 0)$ and gauge a primitive $U(1)_X$. The light-field X charges are the integer PQ lifts used above: $X(F, s, b, S, 10_H, \overline{126}_H, 210_H) = (1, 2, -6, 4, -2, -2, 0)$. Replace the relic-unsafe v19 anomalons by the left-Weyl pairs

$$(P, \overline{P}) \sim (16_1, \overline{16}_{16}), \qquad (Q, \overline{Q}) \sim (16_{14}, \overline{16}_3), \qquad (R, \overline{R}) \sim (16_1, \overline{16}_{-18}). \tag{16}$$

Their accidental-PQ charges are respectively $(1, -1)$, $(-3, 3)$ and $(1, -1)$. Renormalizable masses and decay portals are

$$\Phi^\dagger(y_P P\overline{P} + y_Q Q\overline{Q}) + y_R \Phi R\overline{R} + \lambda_P PF10_H + \lambda_R RF10_H + \lambda_Q \overline{Q}FS^\dagger + \text{h.c.} \tag{17}$$

Every X and PQ charge sum in Eq. (17) vanishes exactly. The two added 16_1 fields share the ordinary-family gauge charges. A generic rank-two Φ mass matrix pairs two linear combinations of the five 16_1 fields with $\overline{16}_{16}$ and $\overline{16}_{-18}$, leaving exactly three chiral light combinations. Thus the labels F, P, R denote a convenient UV basis, not a claim that the heavy–light mass matrix is diagonal.

With $T(16) = 2$ and $\dim 16 = 16$, the continuous anomaly audit is

sector	A_{10^2X}	A_{grav^2X}	A_{X^3}
light fields	-34	-272	-16592
three pairs in Eq. (16)	$+34$	$+272$	$+16592$
total	0	0	0

The cubic entries before the common factor 16 are $4097 + 2771 - 5831 = 1037$. The heavy accidental-PQ mixed anomaly is also zero, so the gauge-anomaly coefficients are unchanged. This does not by itself fix the regular light-fermion current. For a complete singlet-VEV mass block with submatrices (A, B, C, D), where C includes the allowed $SQ\overline{R}$ portal, let $B_\ell(\alpha)$ embed the three canonically normalized light modes. Then

$$Q_{\mathrm{proj}} = B_\ell^\dagger Q_{\mathrm{UV}} B_\ell = \mathbf{1} - 4W, \qquad \mathcal{A}_\alpha = iB_\ell^\dagger \frac{dB_\ell}{d\alpha} = 4W. \tag{18}$$

The sum $Q_{\mathrm{proj}} + \mathcal{A}_\alpha = \mathbf{1}$ is a moving-coordinate identity and is basis dependent; it does not erase the physical projected current. General portal matrices make W non-universal, and misalignment with the SM Yukawa matrices can generate flavour-changing axion currents. A full matching therefore requires the portal tensors and component Yukawa matrices, even though the total anomaly is unchanged.

There is a short minimality proof for an anomalon sector built only from complete pairs, with no Spin(10)-singlet anomalons. If every $16 \oplus \overline{16}$ pair receives a renormalizable mass from one Φ or $\Phi^\dagger$, its two X charges sum to ± 17. Cancellation of the mixed anomaly requires $n_+ - n_- = 1$, hence an odd number of pairs. For one pair, $x + y = 17$ and cubic cancellation require $x^3 + y^3 = 1037$, or $xy = 76$. The roots would satisfy $t^2 - 17t + 76 = 0$, whose discriminant is -15. One pair is impossible, whereas Eq. (16) supplies three; three is therefore minimal under these assumptions. Exact enumeration of portals constructed from the already-present 10_H, S and 210_H finds the displayed triple as the unique three-pair solution. This is not a classification involving arbitrary Spin(10) representations, singlet anomalons or multi-scalar mass operators.

An over-catalogue of every dimension-four-or-lower monomial satisfying Lorentz parity, the Spin(10) centre and exact $U(1)_X$ contains no term that breaks accidental PQ or the light-spectator vector number. The heavy pair numbers are explicitly broken by Eq. (17), as required for decay. Hence PQ remains accidental but no exact anomalon number makes a heavy state stable. PQ is independent of the gauged generator because Φ has $X = 17$ but PQ zero. Writing the canonically normalized phases as (a_Φ, a_S), define

$$D = \sqrt{(17v_\Phi)^2 + (4v_S)^2}, \qquad \widehat{a} = \frac{-4v_S a_\Phi + 17 v_\Phi a_S}{D}, \qquad f_a = \frac{v_S v_\Phi}{D}. \tag{19}$$

The first combination is orthogonal to the gauge direction. For $v_\Phi = 10^{17}\,\mathrm{GeV}$ the correction to $v_S/17$ is -1.10×10^{-12}.

The exact coefficient converting the S phase into $\widehat{a}/f_a$ is

$$\xi = \frac{17v_\Phi^2}{(17v_\Phi)^2 + (4v_S)^2} = 0.0588235294116349 \tag{20}$$

at the frozen v20 point. If the physical projected current is aligned as $Q_{\mathrm{proj}} = \mathbf{1}$ in the SM mass bases, standard axial-basis matching gives the conditional tree benchmark

$$C^0_{u,c,t} = \xi \cos^2\beta, \qquad C^0_{d,s,b} = C_e = \xi \sin^2\beta. \tag{21}$$

Using central low-energy hadronic coefficients [20],

$$\begin{aligned} C_p &= -0.47 + \xi(0.8645\cos^2\beta - 0.437\sin^2\beta), \\ C_n &= -0.02 + \xi(-0.4055\cos^2\beta + 0.833\sin^2\beta). \end{aligned} \tag{22}$$

The constants in Eq. (22) carry correlated hadronic uncertainties of order 0.03 and a precision result requires a declared threshold/RG convention. At the constrained benchmark $\tan\beta = 1.5$ the central values are

$$(C_e, C_p, C_n) = (0.04072398, -0.47214932, 0.00658371). \tag{23}$$

These are not unique numerical v20 predictions. They assume current/Yukawa alignment, while Eq. (18) shows that general portals can change the physical current. Moreover, the corrected flavour analysis must use Takagi diagonalization of the complex-symmetric Majorana matrix and $U_{\rm PMNS} = U_e^\dagger U_\nu$. After this correction the previous $\tan\beta = 1.5$ and high-$\tan\beta$ minima are invalid; the current $v_R = v_S$ profile has no point with $\chi^2 < 30$. Precision matching also requires threshold/RG evolution and correlated hadronic uncertainties.

The large charges make the Abelian running a real constraint. Including all fields above v_Φ gives

$$b_X = \frac{2}{3}\sum_{\rm Weyl} d(R)X^2 + \frac{1}{3}\sum_{\rm scalar} d(R)X^2 = \frac{2}{3}(15840) + \frac{1}{3}(849) = 10843. \tag{24}$$

Requiring the one-loop Landau pole to exceed $M_{\rm Pl} = 2.435\times10^{18}\,\text{GeV}$ at $v_\Phi = 10^{17}\,\text{GeV}$ requires $g_X(v_\Phi) < 0.0478$; $g_X = 0.04$ gives $\Lambda_{\rm LP} = 9.47\times10^{18}\,\text{GeV}$. Complete Spin(10) multiplets do not change one-loop relative unification below their common threshold. Using $T(10,126,210) = (1,35,56)$ [13] and conservatively treating even the real 210_H as complex gives $b_{10} \le 80/3$ above v_Φ. An older diagnostic reset $\alpha_{10}(v_\Phi) = 1/40$ would remain perturbative, but that reset is inconsistent with the model's own spectator-corrected $\alpha_{\rm GUT}^{-1} \simeq 16.81$. Continuous one-loop running from that value is *not* weakly coupled to $M_{\rm Pl}$ under these betas (the conservative envelope hits a Landau pole below $M_{\rm Pl}$). A full two-loop threshold analysis is therefore required before claiming Planck-safe Spin(10) perturbativity.

The decay statement is quantitative. For one open normalized two-body channel with final mass m_ϕ,

$$\Gamma = \frac{|\lambda|^2 M}{32\pi}\Big(1 - \frac{m_\phi^2}{M^2}\Big)^2 \le \frac{|\lambda|^2 M}{32\pi}, \qquad M = \frac{v_\Phi}{\sqrt{2}}. \tag{25}$$

The massless formula is therefore an upper benchmark on the width. At the benchmark, $\lambda = 10^{-8}$ and $m_\phi \to 0$ gives $\tau = 9.36\times10^{-24}$ s, while $\tau < 1$ s in that massless limit requires only $|\lambda| > 3.06\times10^{-20}$. In the explicit Clifford basis, $\sum_{a,j}(C\Gamma^a)_{ij}(C\Gamma^a)^*_{kj} = 10\delta_{ik}$, proving that every component of P and R has a nonzero 10_H channel. The $\overline{Q}FS^\dagger$ singlet contraction is component-diagonal. This removes the exact-stable-relic obstruction; a broken-phase Clebsch and flavour fit is still required for precision widths.

Finally, the portal $|\Phi|^2|S|^2$ must be small enough to preserve $v_\Phi \gg v_S$. These are explicit perturbativity and hierarchy assumptions, so this is a field-theory EFT valid only below its first strong-coupling or UV-completion threshold. The present running does not establish a weakly coupled completion to the Planck cutoff, let alone an ultimate quantum-gravity construction.

6 Heavy-threshold closure and full loop kernels

Heavy fields can invalidate naive axion-quality power counting [11], so the operator search was repeated with all fields in Eq. (16) and their conjugates. Because the decay vertices break the heavy pair numbers, only the exact light-spectator number V_s is imposed. The resulting minimum-dimension frontier has no PQ-sensitive, $V_s = 0$ closure through $P = 7$. At $P = 8$ the necessary-condition bound is saturated by actual Spin(10) invariants:

$$\begin{aligned}
\widetilde{\mathcal{O}}_5 &= \frac{(S^\dagger)^2(Qb)_1}{M_{\rm Pl}}, && (Q_{\rm PQ}, V_s, P) = (-17, -1, 1),\\
\widetilde{\mathcal{O}}_8 &= \frac{(S^\dagger)^2[(ss)_{10}]^2}{M_{\rm Pl}^4}, && (Q_{\rm PQ}, V_s, P) = (0, +4, 4).
\end{aligned} \tag{26}$$

The combination $4\widetilde{\mathcal{O}}_5 + \widetilde{\mathcal{O}}_8$ closes with four chirality-flipping s–b propagators generated by Ssb, four Q–$\overline{Q}$ propagators generated by $\Phi^\dagger Q\overline{Q}$, four $\overline{Q}FS^\dagger$ decay vertices and two conjugate ordinary $FF10_H$ Yukawas. The left–left propagators carry the conjugate mass spurions, $S^\dagger$ and Φ, respectively. Counting the five Planck vertices, four decay vertices and two ordinary Yukawas gives a connected graph with 12 internal fermion lines, 11 vertices and two loops. It produces

$$\Phi^4 (S^\dagger)^{18} (10_H^\dagger)^2, \qquad (X, Q_{\rm PQ}, P) = (0, -68, 8). \tag{27}$$

The explicit Clifford representation gives $(T^a)^*_{ij}(T^b)_{ij} = 16\delta^{ab}$ and the inherited four-bilinear two-component contraction, including both kinetic numerators, equals -2 per factorized loop (hence 4 for their product), so this saturation graph is nonzero.

The original v17 two-loop graph also survives, but its four $\mathcal{O}_6^-$ portals each carry $X = -17$. Exact gauge invariance replaces each by $\Phi\mathcal{O}_6^-/M_{\rm Pl}$; the graph is therefore $P = 16$. The full zero-momentum integral can be written without dimensional estimates. For degenerate spectator mass M_s and a family chirality mass m,

$$\begin{aligned} I_3(M_s, m) &= \int \frac{d^4p_E}{(2\pi)^4} \frac{1}{(p^2 + M_s^2)^2 (p^2 + m^2)} \\ &= \frac{1}{16\pi^2 M_s^2} \frac{1 - r + r\ln r}{(1-r)^2}, \qquad r = \frac{m^2}{M_s^2}, \end{aligned} \tag{28}$$

$$J(M_s, m) = M_s^2 m\, I_3(M_s, m),$$

$$|A_{2\ell}^{\rm EFT}| = |C_{2\ell}| \frac{s_0^{14}}{M_{\rm Pl}^{12}} J(M_s, m_1) J(M_s, m_2), \qquad s_0 = \frac{v_S}{\sqrt{2}}. \tag{29}$$

For unequal masses the code evaluates the symmetric divided difference

$$I_3 = \frac{1}{16\pi^2} \sum_x \frac{x \ln x}{\prod_{y \neq x}(x - y)}. \tag{30}$$

Equation (29) is finite and factorizes because the zero-momentum four-fermion vertex leaves two independent loop momenta. The normalized coefficient $C_{2\ell}$ contains the unknown Planck Wilson tensors, flavour contractions, the nonzero group/Lorentz normalization and RG matching; the EFT cannot predict it.

For the $P = 8$ graph, each of the two factorized loops has the independently evaluated kernel

$$K(M_H, M_s, m_f) = M_H^2 M_s^2 \int \frac{d^4p_E}{(2\pi)^4} \frac{p^2}{(p^2 + M_H^2)^2 (p^2 + M_s^2)^2 (p^2 + m_f^2)^2}. \tag{31}$$

High-precision repeated-pole partial fractions retain all three masses and are independently checked by log-coordinate quadrature. The one-sided amplitude is

$$|A_8| = |C_8| \frac{s_0^{14} h^2}{M_{\rm Pl}^8} K(M_H, M_s, m_f)^2. \tag{32}$$

Here C_8 contains the four decay Yukawas, two family Yukawas and the normalized Wilson, group, flavour and RG contraction. With $M_s = v_S$, $M_H = v_\Phi/\sqrt{2}$, $m_f = h = 246\,\text{GeV}$ and $v_\Phi = 10^{17}\,\text{GeV}$, $16\pi^2 M_H^2 K = 0.9999999965$. The results per unit normalized coefficient are

term	$\lvert A\rvert/\chi$	worst phase $2\lvert A\rvert/\chi$
v17 two-loop EFT graph, before Φ dressings	2.15×10^{-53}	4.30×10^{-53}
same graph in $U(1)_X$ ($P = 16$)	1.53×10^{-59}	3.06×10^{-59}
decay-threshold graph ($P = 8$)	3.02×10^{-47}	6.04×10^{-47}
direct $\Phi^4 (S^\dagger)^{17}/M_{\rm Pl}^{17}$	2.30×10^{-43}	4.60×10^{-43}

The direct dimension-21 scalar operator dominates these explicitly computed terms and still lies 2.17×10^{32} below the 10^{-10} criterion for a unit coefficient. This numerical margin is conditional on order-one normalized Wilson contractions; it is not a measurement or a derivation of quantum-gravity coefficients.

An extra vector-like spectator $\mathbb{Z}_3$ is neither needed nor used. Indeed $\mathcal{O}_{10}$ and $\mathcal{O}_{12}$ survive it, falsifying the earlier claim that all fermionic breakers move to dimension 19. Such an exact triality would also make the lightest triality-charged state stable unless further structure were added. The rank-six rephasing diagnostic in Appendix A remains correct but is not the quality proof. The continuous origin is supplied in Section 5; its cosmic-string spectrum remains open.

Adversarial merge audit of the alternate anomalon lift

The alternate charges $16_1 \oplus \overline{16}_{16}$ and singlets $(33, -16, 31, -48)$ do cancel the three continuous anomaly sums exactly. They do not, however, establish the advertised quality theorem. A physical axion first requires a global accidental PQ independent of the gauged generator; one possible repair assigns $Q_{\rm PQ}(16_1, \overline{16}_{16}, \Phi) = (1, -1, 0)$. With that assignment the following actual 10-channel invariants are allowed:

$$
\begin{aligned}
cal O_6^A &= \frac{(S^\dagger)^2(\overline{16}_{16}\,\overline{16}_{s,-6})_{10}10_H}{M_{\rm Pl}^2}, && (Q_{\rm PQ}, V_s, P) = (-17, -1, 2), \\
cal O_8^s &= \frac{(S^\dagger)^2[(16_{s,2}16_{s,2})_{10}]^2}{M_{\rm Pl}^4}, && (Q_{\rm PQ}, V_s, P) = (0, +4, 4).
\end{aligned} \tag{33}
$$

Hence $4\mathcal{O}_6^A + \mathcal{O}_8^s$ is a $(Q_{\rm PQ}, V_s, P) = (-68, 0, 12)$ closure. The proposed $16_1 16_F 10_H$ vertex breaks anomalon number, and four anomalon-mass propagators, four such vertices and two ordinary $16_F 16_F 10_H$ vertices provide a connected two-loop topology for the remaining fermions. An end-to-end Grassmann contraction of that alternate topology is still required. The scalar-only dimension-21 scan therefore misses the relevant fermionic threshold; this does not prove that the alternate model is unsafe, but it does mean its published quality bound and loop amplitude are incomplete.

The same audit finds only the two Φ mass Yukawas for the four heavy singlets, not a renormalizable portal to light matter. It also gives $b_X = 8263$ and $g_X(M_{\rm GUT}) < 0.0417$ for a Landau pole above $M_{\rm Pl}$. Finally, applying the exact factorized kernel to the *old* graph at the alternate inputs gives $|A|/\chi = 1.85 \times 10^{-64}$ after four dressings, rather than 2.37×10^{-62}; the ratio is the earlier 128-fold vev-normalization envelope. These are the reasons the alternate anomalon sector is not merged, while its charge-general minimality proof is retained.

7 Status of cosmology, experiment, and proton decay

Cosmology (conditional). The v20 anomalons carry no exact heavy number and decay through Eq. (17); unlike v19, avoiding a stable heavy relic does not require a low-reheat postulate. This statement does not replace a Boltzmann calculation if the $U(1)_X$ sector is thermally restored. Pre-inflationary: $\theta_i = 2.91$ on the anharmonic tail [15], $H_I \lesssim 9 \times 10^5\,\text{GeV}$ (UV-conditional [16]), corresponding to $r < 1.34 \times 10^{-17}$ for $A_s = 2.1 \times 10^{-9}$. A primordial B-mode detection above that ceiling rejects this pre-inflationary all-dark-matter branch, not the post-inflationary branch. The latter is conditional on the $(\ell, n) = (13, -3)$ string's existence and production [14]. If a standard network applied, current simulations disagree on the full-DM mass: Benabou et al. find 45–65 μeV before wall collapse, with wall effects potentially extending to $\sim 300\,\mu$eV [17]; Saikawa et al. report 95–450 μeV [18]. The benchmark lies inside the latter and inside the extended range of the former; the nonstandard gauged network makes all such intervals indicative.

Experiment. No dark-matter haloscope has scanned 153.5 μeV: the recent DALI results sit at 28.5 μeV [22] (their reported sensitivity is disputed [23]), and no existing experiment reaches the predicted coupling there; CAST constrains the mass at $5.8 \times 10^{-11}\,\text{GeV}^{-1}$ [25], a factor

2 479 above. Direct tests belong to MADMAX [26] and plasma-haloscope [27] programmes; the ORGAN design range also overlaps the target [28]. Their design ranges are not completed scans or guaranteed DFSZ sensitivity. The benchmark location is 36.62–37.60 GHz, while the Galactic signal width is only about 34 kHz for $Q_{\rm halo} = 1.1\times10^6$. A null scan of the full location band below $g_{a\gamma\gamma} = 2.335 \times 10^{-14}\,\mathrm{GeV}^{-1}$ would reject the all-dark-matter benchmark combination ($v_S{=}M_I$, $y_s{=}\sqrt{2}$), not the anomaly theorem itself.

Proton decay (prior-region statements). Under the diagnostic priors of Table 1, the median lifetime is 1.1×10^{35} yr; 35 per cent of sampled volume lies below the Super-K cutoff 2.4×10^{34} yr [29] and a further 14 per cent below the Hyper-K reach 1×10^{35} yr [30]; a Hyper-K null would exclude only that additional subregion.

Table 1: Diagnostic priors of the proton Monte Carlo (seed 20260801, 10^5 trials; spectator shift applied per trial with $y_s = \sqrt{2}$).

$\alpha_s(M_Z)$	$\mathcal{N}(0.1179,\, 0.0009)$
$\alpha_1^{-1},\ \alpha_2^{-1}$	$\mathcal{N}(59.02, 0.02)$, $\mathcal{N}(29.57, 0.02)$
threshold	$\Delta_4 = \delta \sim \mathcal{U}(-1, 1)$, $\Delta_{2R} = -\frac{2}{3}\delta$, $\Delta_{2L} = 0$
$M_X/M_{\rm GUT}$	LogU(0.5, 2)
A_R	$\mathcal{U}(2.0,\, 3.2)$
W_0 [GeV^2]	$\mathcal{N}(0.1095,\, 0.011)$

8 Novelty and conclusions

The defensible contribution now has five layers: (i) the charge-independent five-pair lower bound, plus residue uniqueness in the identical-pair ansatz, for this particular $\mathbb{Z}_{17}$ extension; (ii) the discrete-EFT $P = 12$ multi-spurion lower bound and its explicit Spin(10) graph; (iii) the anomaly-free $U(1)_X \to \mathbb{Z}_{17}$ field-theory embedding of Section 5; (iv) the one-pair no-go and minimal three-pair decay-safe completion; and (v) the heavy-inclusive $P = 8$ closure audit plus its repeated-pole two-loop integral in Section 6. These are narrow model-building and matching results within the established discrete-gauge axion literature [6, 7, 8, 9], not a derivation of the full ERT parameter point from one scale and not experimental evidence for an axion.

The analytic charge ledger for the continuous embedding passes its stated internal tests: all local anomalies vanish, renormalizable PQ remains accidental, the axion survives the gauge projection, and the explicitly evaluated unit-coefficient quality terms are far below the neutron-EDM requirement. Every anomalon pair now has a renormalizable ordinary-matter portal, and a generic mass matrix leaves three chiral families. The completion also changes the EFT interpretation: the old two-loop graph is $P = 16$, while the decay-enabled heavy matter opens a $P = 8$ two-loop closure. Its exact benchmark shift is nevertheless four orders below the direct dimension-21 scalar term, which dominates the calculated terms. Unknown Wilson/flavour/Yukawa tensors remain parameters, so "full amplitude" here means the complete finite momentum kernel and mass dependence for the stated graph, not a coefficient-free prediction. It is not an external implementation certificate. The repository model file now uses native non-supersymmetric SARAH syntax, includes the manuscript's gauged $U(1)_X$, and passes static catalogue, charge, Lagrangian, filter, and hash-bound input-manifest checks. A SARAH 4.15.3 run on Wolfram Engine 15.0, with six SO(N) defects of that release replaced by exact routines on its bundled Susyno engine, registers all 19 Lagrangian terms as SO(10), $U(1)_X$ and $\mathbb{Z}_{17}$ invariant with vanishing anomaly sums, and rejects a forbidden $210_H 10_H^\dagger 10_H$ term and a shifted X charge. This attestation verifies the model contract (fields, charges, Yukawa and mass terms, singlet quartics), not the non-singlet quartics of the census below, and it is not by itself a scalar-vacuum proof. The exact-X renormalizable scalar census contains 44 tensor directions and 51 real parameters. Its dense derivative audit passes. An exact Gaussian-integer projector identity proves that the selected-vacuum 54 and

$\overline{1050}$ projector-gradient columns vanish, and a separate exact integer/rational calculation proves the mixed Φ–$\overline{126}$ 210 column also vanishes. An exact nonzero 13×13 compiler-bound minor and an exact full-row factorization prove stationarity rank 13 and nullity 38. Normalized floating-point SVD agrees but is used only as a diagnostic; an exact 38-vector parameter-nullspace basis is not yet part of the G3 solver contract. Independently, an exact Gaussian-integer tangent certificate proves that the broken SO(10) plus gauged $U(1)_X$ orbit has rank 37. Its gauge quotient is therefore 449-dimensional and includes the physical axion. The independent global-PQ tangent raises the full symmetry orbit rank to 38; quotienting that flat orbit for Hessian positivity leaves a 448-dimensional massive/transverse space. PQ is not gauge-eaten. These ranks hold at the declared $p+\delta$ point ($p = e_{6789}$, $z_k = e_{2k-2} + ie_{2k-1}$), whose $\delta = z_1 \wedge z_2 \wedge z_3 \wedge (e_{67} + e_{89})$ is the $Y = -1$ member of the $(\overline{10}, 1, 3)$, not the SM singlet; at the Pati–Salam G3 witness they are 34 and 35 (quotients 452 and 451; G4). The neutral Φ_{210} projector P_{24} is exactly symmetric, idempotent, and rank 24. An exact normalized stationary witness with coefficients $(10, 1, -1/4)$ has vanishing dense gradient and $\mathrm{tr}(P_{24}H) = 288$. The former normalized-SVD stationary constraints reject this witness, so the associated finite-cut, common-kernel, block-SDP, and negative-trace-LP results are invalidated. The H_6 radial coordinate of that reference point is also not an exact flat: writing its canonical real coordinate as $t = \sqrt{2}h$, the exact stationary combination $c_{O06} = -t^2$, $c_{O36,1} = 10$ has vanishing first derivative and curvature $2t^2 = 4h^2 > 0$. Adding the independent normalized P_{24} witness preserves this result and the coupling bound.

Using the corrected 11-plus-2 stationary constraints directly on the raw orthonormal 448-dimensional massive/transverse quotient, an opt-in float64 recomputation gives common-Gram rank 448 and nullity zero. The former apparent 135-dimensional common flat subspace is reproduced only after a reference-derived diagonal congruence with condition ratio approximately 1.20×10^8 and is invalidated as a conditioning artifact. This numerical full-rank result is not a Hessian-positivity certificate.

The corrected search nevertheless produces a new constructive frontier. A sparse exact-X coefficient vector uses 27 of the 51 real parameters, obeys $\|c\|_\infty = 73/8 < 4\pi$, and has $J_0 = -21/200$. It is therefore outside the former $J_0 = +1$ slice, showing that this anchor was not without loss of generality. The vector has a manifest sum-of-squares structure. In particular, a direct Gaussian-integer tensor calculation together with exact rational Casimir projectors proves the six-channel recoupling

$$\|\mathcal{M}(\Phi)\Sigma\|^2 = 40I_1 + 72I_{45} + 28I_{210} - 8I_{770} - 12I_{5940} + 12I_{8910}. \tag{34}$$

A nonsingular rational six-witness evaluation fixes these coefficients uniquely. A separate source-bound expansion of all 27 nonzero couplings proves that the complete potential is a sum of nonnegative squares. It also proves exact stationarity of the selected vacuum, including all 252 real $\overline{126}$ coordinates. Thus boundedness below and selected-vacuum stationarity are exact statements, not numerical scans.

For the $\Phi_{210} + \delta$ Hessian, direct Gaussian-integer and rational tensor contraction followed by exact $\mathbb{Q}(\sqrt{2})$ component arithmetic gives $\mathrm{rank}\, K = 278$, $\mathrm{rank}\, H_\Phi = 186$, and $\mathrm{rank}(H_\Phi + K) = 429$ with nullity 33. The proof assembles every entry from integer/Gaussian-integer tensors and exact projector fractions; the float64 live-source comparison is only a regression check. The exact $P + \delta$ gauge-orbit matrix has rank 33. After quotienting that kernel, an explicit 26×24 integer Jacobian for the nonnegative $H/S/\Phi_{17}$ constraints has exact rank 19. The five remaining added directions combine with the 33 core gauge tangents to give precisely the 38 SO(10), gauged-$U(1)_X$, and global-PQ symmetry tangents. Hence the full Hessian has rank 448 and is positive on all transverse directions.

Combining exact stationarity, complete-potential BFB, and transverse Hessian positivity proves that the selected symmetry orbit is a strict local minimum. The final exact global-gap test is decisively negative. In canonical $\overline{126}$ coordinates there is a Gaussian- integer field direction with normalized projector fractions $(I_{54}, I_{1050}, I_{2772}, I_{4125}) = (0, 0, 1/2, 1/2)$. It annihilates both mixed squares at P and, after choosing $N_\Sigma = (100/99)r^2$, has energy below the selected orbit by

$25r^4/19008 > 0$. The selected orbit is therefore not the global vacuum and this 27-parameter candidate is rejected for G3, which does not exclude the theory: the fixed-P branch with $\Sigma \propto \delta$ obeys the exact relation $\Delta V = -m_\perp^2/8$, and its lower replacement has the wrong stabilizer. A distinct $p : a : \omega = 1 : 1 : 1$ SU(5)-singlet branch with $\Sigma \propto \delta$ has an explicit global $\Phi + \Sigma$ SOS and exact rank/nullity 429/33, but its stabilizer $SU(3)_c \times SU(2)_L \times U(1)_{T_{3R}}$ is not the Standard Model. With the chiral vector $H = (e_6 + ie_7)/\sqrt{2}$, its 28-parameter full-field extension is exactly stationary and BFB, has symmetry ranks 36/37/38 and a live positive 448-dimensional quotient with minimum eigenvalue 0.00484459. An exact source-bound rational-lattice block-LDL certificate proves full Hessian rank/nullity 448/38, positive semidefiniteness, and a kernel equal to the symmetry orbit. Being non-SM (δ breaks hypercharge; $|\langle H\rangle|/|\langle\Phi\rangle| = 1$), this point can never close G3. The literal single-orbit four-form lemma is false: F and $-F$ obey both projector equations but are separated by the cubic invariant $\mathrm{Tr}(A_\Phi^3) = \pm 60$ for $\pm F_0$, $|F_0|^2 = 10$. Exact mixed rank 252/252 excludes the coupled $-F$ equality branch. An exact implicit-function and equivariance argument proves both signed orbits are isolated local components of the full projector-zero set. The complete 16-real-dimensional SU(3)-fixed subspace is classified exactly and contains no further orbit. At fixed $\Phi = F$ an exact off-kernel estimate proves the full $\beta = 1/20$ gap for arbitrary H and Σ, with equality only on the selected SU(5) flag orbit. The strongest all-zero-residual route is also removed exactly. On the pure-δ, maximally negative-current stratum the live integral affine system has rank/nullity 168/42 and its integer kernel splits as $35 + 7$. Exact square completion gives a Φ radial-plus-I_{54} lower bound $1/141$ and a worst radial/current value $-7001/995000$, hence the positive margin

$$\frac{1}{141} - \frac{7001}{995000} = \frac{7859}{140295000} > 0. \tag{35}$$

An exact source-bound certificate strengthens this: it retains all shifted Φ–Σ and chiral Φ–H residuals and covers every real Φ and all nonnegative radial variables on the same maximally negative pure-δ sector. A single exact 4125-projector covariant, a rational Schur-complement bound, and a three-region radial completion prove the sharp restricted gap

$$\Delta V_{\text{pure-}\delta} \geq \frac{1}{5000}, \tag{36}$$

with equality at $u = 1$, $v = 0$ for a signed Kähler-square representative. The corrected v21 theorem subsumes the earlier SU(3) regression. At fixed $H = h_-$ and $\Sigma = q/4$, a source-reconstructed exact positive-Gram identity proves $p(t, \Phi) > 0$ away from the homogeneous origin for every real Φ_{210}. Hence at $t = 1$, $A(\Phi) > 3/200$ and the p-zero set is empty. This theorem does not vary H or Σ and does not classify the full Hessian or close G3. At that endpoint the common continuous stabilizer is exactly SU(4): the joint kernel has dimension 15, the integral Lie brackets close, and all 15 skew actions on Φ_{210} are source-bound. Exact intertwiners decompose the complexified 210 into 25 carriers, which common lowering words align into an exact rank-210 basis with physical conjugation and real-form maps. An exact 5952×551 constraint system has rational rank 506 and nullity 45; its integral nullspace is a complete basis of 45 real symmetric quadratic forms commuting with all 15 live actions. Thus $\dim\big(\mathrm{Sym}^2(\Phi_{210})\big)^{\mathrm{SU}(4)} = 45$. The augmented homogeneous census is also exact: real dimension 22366, with 35 complex isotypic types spanning 824 irreducible copies and 22 real Schur blocks: 9 real-symmetric and 13 complex-Hermitian. Their multiplicity matrices contain 19594 real Schur parameters, graded $(1, 4, 90, 1414, 18085)$, while the invariant quartic target has 6585 rows, graded $(1, 4, 45, 478, 6057)$. A universal grade-preserving rational polarization section proves the abstract invariant multiplication map surjective, with total rank 6585 and kernel dimension 13009. The cubic Schur interface is coordinate-exact: exact highest-weight carriers give all 540 required copies of the ten relevant $\mathrm{Sym}^2(\Phi_{210})$ irreducible families and, after physical realification, all 1414 cubic cross variables in the seven nonzero real/Hermitian blocks; their symmetric-Gram multiplication map is an integer 478×1414 matrix of exact rank 478 and kernel dimension 936. Thus every abstract invariant cubic coefficient vector has a linear preimage. This is not a PSD result: the interface's reserved zero vector is not the physical G3 target and certifies no vanishing physical cubic right-hand side. The quartic Schur interface is coordinate-exact

too. All 35 complex carrier families, spanning 798 irreducible copies and 22155 carrier dimensions, and all 22 real block pairings give a sparse integer 6057×18085 map with 115641 nonzero entries, exact rank 6057, and kernel dimension 12028. The legacy v20 assembled physical target is rejected (structural provenance only; its public entrypoints fail closed). The corrected standard positive-Gram map has shape 6585×19594, denominator 256, and 138550 nonzero entries; its ordered-spectral target has denominator 576000 and 512 nonzero entries. All 6585 rational coefficient equalities hold, and the exact primal has 22 strictly positive real/Hermitian blocks with 824 positive rational LDL pivots. With the complete invertible carrier census this proves the arbitrary-real-Φ_{210} statement at the fixed $H = h_-$, $\Sigma = q/4$ endpoint only. Global Σ and general or full H remain open for this non-SM point, which can never close G3. Historical finite-cut and SDP solvers are non-certifying and unused in the local-minimum proof. G3 is closed only on the Pati–Salam benchmark family: $(\Phi, \Sigma) \propto (p, z_1 \wedge \cdots \wedge z_5)$ is the exact SM-preserving global vacuum of a distinct, retuned 27-parameter benchmark of the declared exact-X potential with the light-doublet deformation $\epsilon H^\dagger H$, $\epsilon > 0$ (unique modulo $\mathrm{SO}(10) \times \mathrm{U}(1)_X \times$ accidental $\mathrm{U}(1)_{\mathrm{PQ}}$; exact Hessian at $r_0 = M_I/M_{\mathrm{GUT}} = 1/5$, $|\Phi_{17}| = M_{\mathrm{GUT}}$: rank/nullity 451/35, kernel the symmetry orbit; six cited classical theorems and six hand-argued steps). The doublet–triplet and M_I relations are tuned. G4 is closed at this witness at tree level: the Hessian on the 452-dimensional gauge quotient has inertia 451/1/0, its only zero mode the axion, which is massless only for this renormalizable PQ-neutral potential. No internal candidate is approved: that repository tier is held until G7 closes, since for the declared benchmark ($5 \times (16_s \oplus \overline{16}_s)$, $v_S = M_s = M_I$) three independent two-loop codes (research runs, not yet part of this release) find no perturbative unification. G6 is open and G7–G8 are blocked ($\overline{126}$ colour states below M_I, no electroweak breaking, RG-anchor content, quartic 127/64, proton-decay mediators), and on this branch the renormalizable Yukawas give CKM = 1 and $m_t = m_b$, which data exclude. This work neither validates nor excludes the complete model.

The v16–v20 audit also records negative results: the v15 dimension-9 one-insertion potential and its $\mathbb{Z}_3$ cure were incorrect, and the alternate v18 scalar-only UV-quality proof is incomplete. V19's exact stable-anomalon caveat is resolved rather than hidden: its charge set is replaced. The remaining scientific tasks are an independent representation-theory and diagrammatic review, broken-phase Clebsch and flavour fits, RG evolution of the Wilson tensors, radiatively stable v_Φ/v_S, doublet–triplet and M_I hierarchies, a thermal-restoration calculation, an ultimate quantum-gravity origin, and the nonstandard string-network calculation.

A Reproduction formulas

The rephasing diagnostic referenced in Section 4 is

$$M = \begin{pmatrix} 0 & 2 & 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 1 \\ 2 & 0 & 2 & 2 & 0 & 0 \\ -3 & 1 & 0 & 0 & 0 & 1 \\ 4 & 1 & 1 & 0 & 1 & 0 \\ -2 & 1 & 0 & 0 & 1 & 2 \\ -2 & 0 & 0 & 0 & 4 & 0 \end{pmatrix} \tag{37}$$

over $(X_S, X_F, X_{10}, X_{\overline{126}}, X_a, X_b)$, rows: the two Yukawas, spectator mass, ERT lock, $(S^\dagger)^3 16_F \overline{16}_s$, $S^4 16_F 16_s 10_H$, $(S^\dagger)^2 (16_F \overline{16}_s)(16_s \overline{16}_s)$, $(S^\dagger)^2 (16_s 16_s)_{10} (16_s 16_s)_{10}$. The proton width used in the Monte Carlo is

$$\Gamma(p \to e^+ \pi^0) = \frac{m_p}{32\pi} \left[1 - \frac{m_\pi^2}{m_p^2}\right]^2 \left(\frac{4\pi\alpha_{\mathrm{GUT}}}{M_X^2}\right)^2 A_R^2 \, |W_0|^2 \left[1 + (1 + |V_{ud}|^2)^2\right], \tag{38}$$

with W_0 from lattice QCD [21] and $\tau_p = \hbar/\Gamma$.

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